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Design and Implementation of a Data Warehouse and Business Intelligence Solution

Berka Financial Dataset

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Table of Contents

Task 1: Dataset Selection and Business Scenario Identification 1
1.1 Business Domain 1
1.2 Purpose of the System 1
1.3 Business Problem 1
1.4 Dataset Source 1
1.5 Dataset Format 1
1.6 Number of Records 2
1.7 Main Attributes 2
1.8 Dataset Suitability for DWBI 2

Task 2: Data Source Identification and Preparation 3
2.1 Data Sources 3
2.2 Relationships Between Sources 3
2.3 Data Preparation 4
2.3.1 Importing the source files into SQL Server 5
2.3.2 Checking record counts and column structures 5
2.3.3 Checking missing values in key identifier fields 6
2.3.4 Checking duplicate values in key fields 6
2.3.5 Validating data types and date formats 6
2.3.6 Checking relationships between related tables 7
2.3.7 Identifying invalid or inconsistent values 7
2.3.8 Standardizing values where required 7
2.3.9 Preparing clean data for the Data Warehouse 8
2.4 Source Integration 8

Task 3: Data Warehouse Architecture Design 9
3.1 Architecture Diagram 9
3.2 Data Source Layer 9
3.3 Data Integration Layer 9
3.4 Storage Layer 10
3.4.1 Staging Area 10
3.4.2 Enterprise Data Warehouse 11
3.4.3 Data Mart 11
3.5 OLAP Layer 11
3.6 Presentation Layer 13

Task 4: Dimensional Data Warehouse Design and Implementation 14
4.1 Business Process 14
4.2 Grain 14
4.3 Fact Table & Measures 15
4.4 Dimension Tables 17
4.4.1 DimDate 17

4.4.2 DimAccount 17
4.4.3 DimCustomer 18
4.4.4 DimDistrict 18
4.4.5 DimTransactionType 19
4.4.6 DimOperation 19
4.5 Foreign Keys & Relationships 19
4.6 Schema Selection 21
4.7 Dimension Hierarchies 22
4.8 Table Relationships 23
4.9 Table Descriptions 25
4.9.1 FactTransaction 25
4.9.2 DimDate 26
4.9.3 DimAccount 26
4.9.4 DimCustomer 27
4.9.5 DimDistrict 27
4.10 Design Assumptions 28
Task 5: ETL Process Development 30
5.1 ETL Overview 30
5.1.1 ETL Implementation Using SQL Server 31
5.2 Extract 32
5.3 Transform 32
5.3.1 Date Conversion 32
5.3.2 Customer Birth Date Transformation 33
5.3.3 Data Type Conversion 33
5.3.4 Missing Value Handling 33
5.3.5 Data Integration 33
5.3.6 Data Validation 34
5.4 Load 35
5.4.1 Loading Dimension Tables 35
5.4.2 Loading Fact Table 35
5.4.3 Surrogate Key Mapping 36
5.4.4 Load Validation 36
5.5 ETL Validation and Results 36
5.5.1 Source-to-Warehouse Row Count Validation 36
5.5.2 Foreign Key Validation 36
5.5.3 Duplicate Transaction Validation 37
5.5.4 Transaction Completeness 37
5.5.5 Missing Operation Validation 37
5.5.6 ETL Validation Summary 37
Task 6: Data Mart Development 39
6.1 Data Mart Overview 39

6.2 Data Mart Implementation 39
6.3 Purpose, Target Users and Analytical Benefits 41
6.3.1 Purpose of the Data Mart 41
6.3.2 Target Users 41
6.3.3 Analytical Benefits 42
Task 7: OLAP Analysis and Business Intelligence Dashboard Development 43
7.1 Introduction 43
7.2 OLAP Operations 43
7.2.1 Roll-up 43
7.2.2 Drill-down 43
7.2.3 Slice 43
7.2.4 Dice 43
7.2.5 Pivot 43
7.3 Power BI Dashboard Development 43
7.4 Dashboard Analysis 45
7.5 Conclusion 46
Task 8: Business Insights and Recommendations 47
8.1 Introduction 47
8.2 Business Insights 47
8.2.1 Yearly Transaction Amount Growth 47
8.2.2 Monthly Transaction Volume 47
8.2.3 Customer Gender Comparison 47
8.2.4 Regional Transaction Differences 47
8.2.5 Transaction Type Patterns 48
8.2.6 District-Level Transaction Activity 48
8.2.7 Banking Operation Amounts 48
8.3 Overall Recommendations 48
8.4 Conclusion 48
Use of AI Tools and Acknowledgement 50
References 51

Task 1: Dataset Selection and Business Scenario Identification

1.1 Business Domain

Banking and Financial Services

The selected dataset represents a banking environment that provides services to private customers. The dataset contains information related to customers, bank accounts, transactions, loans, credit cards, payment orders, and district-level demographic information.

1.2 Purpose of the System

The purpose of the proposed system is to integrate banking data into a Data Warehouse to support historical analysis and business intelligence. The system will help management analyse customer activity, account transactions, loans, credit cards, and regional patterns.

1.3 Business Problem

The bank needs to analyse customer activity, account transactions, loans, credit cards, and regional patterns using data stored across multiple operational tables. Analysing these tables separately makes it difficult to obtain a consistent historical view of banking activities. The proposed Data Warehouse will integrate these data sources and provide a centralized environment for analytical reporting and business decision-making.

1.4 Dataset Source

The selected dataset is the **Berka Financial Dataset from the PKDD'99 Discovery Challenge**. It contains anonymized financial data from a banking environment. The original dataset description includes information about customers, accounts, transactions, loans, credit cards, payment orders, and demographic data.

1.5 Dataset Format

The dataset is provided as multiple .ASC files representing relational data tables. The files can be imported into Microsoft SQL Server and used as the operational source for the Data Warehouse.

1.6 Number of Records

Table 1.1: Number of records in each source table

Source Table	Number of Records
ACCOUNTS	4,500
CLIENTS	5,369
DISP	5,369
ORDERS	6,471
TRANS	1,056,320
LOAN	682
CARDS	892
DISTRICT	77

The transaction table contains 1,056,320 records, providing a substantial volume of data for analytical processing.

1.7 Main Attributes

The dataset contains customer, account, transaction, loan, credit card, payment order, and demographic attributes. Important transaction attributes include transaction date, transaction type, operation, amount, and account balance. Account and customer identifiers provide relationships between the operational tables. District attributes provide additional demographic information for analysis

1.8 Dataset Suitability for DWBI

The Berka dataset is suitable for developing a Data Warehouse and Business Intelligence solution because it contains multiple related operational tables and a large number of transaction records. The data supports the identification of business processes, facts, measures, and dimensions. It also provides information for analysing transactions, customer activity, loans, credit cards, accounts, and geographical patterns. Therefore, the dataset provides a suitable foundation for ETL processing, dimensional modelling, OLAP analysis, and BI reporting.

Task 2: Data Source Identification and Preparation

2.1 Data Sources

The Berka banking dataset was obtained from the PKDD'99 Discovery Challenge and contains banking information stored in multiple .ASC files. The source files represent eight relational tables: ACCOUNTS, CLIENTS, DISP, ORDERS, TRANS, LOAN, CARDS, and DISTRICT.

The source data was initially imported into Microsoft SQL Server and stored in the Berka_DWBI database. Two CSV files were also included as supporting data files for the assignment. The imported source tables were used for data profiling, preprocessing, and validation before transferring the prepared data into the dimensional Data Warehouse.

SQL scripts were saved separately to document the source data preparation, validation, and subsequent data transfer steps.

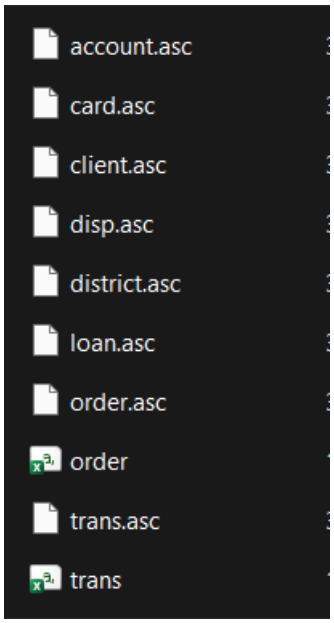


Figure 2.1: Berka source data files provided for the project

2.2 Relationships Between Sources

2.2 Relationships Between Sources

The Berka dataset consists of related banking tables connected through common identifiers. The DISP table links clients and accounts using client_id and account_id. The TRANS, ORDERS, and LOAN tables are associated with bank accounts through account_id. The CARDS table is linked to the DISP table through disp_id. Customer and account district information is connected to the DISTRICT table using district_id.

These relationships allow data from the different source tables to be integrated while maintaining the connections between customers, accounts, and banking activities.

The relationships were checked using SQL queries after importing the source data. Orphan records were not identified in the relationships that were checked.

2.3 Data Preparation

After importing the eight source tables into the Berka_DWBI database, data profiling and validation were performed before transferring the data to the Data Warehouse. The preparation process focused on checking record counts, missing values in key identifier fields, duplicate records, data types, date formats, relationships between tables, invalid or inconsistent values, and unnecessary leading or trailing spaces.

SQL queries were used to examine the source data and identify potential data quality issues. The validation results were reviewed to determine whether the data was suitable for subsequent dimensional modelling and loading into the Berka_DW Data Warehouse.

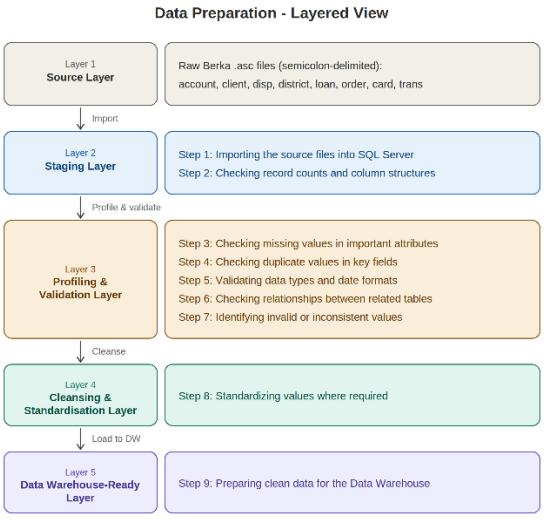


Figure 2.2: Layered view of the data preparation process

2.3.1 Importing the source files into SQL Server

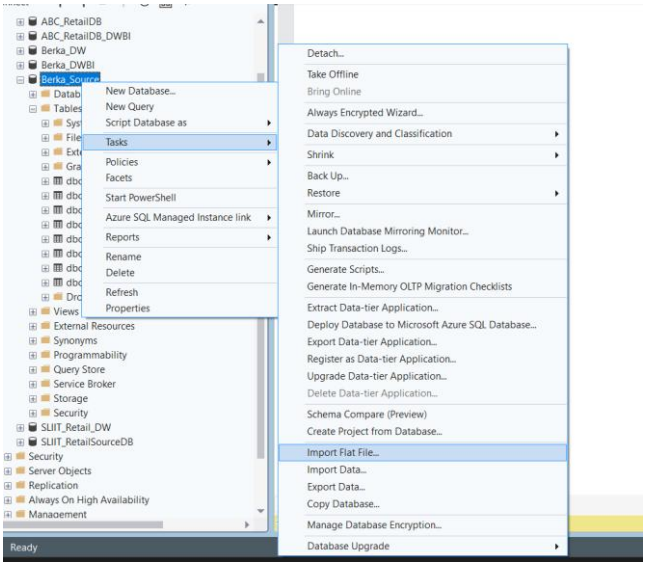


Figure 2.3: Importing the Berka source files into SQL Server using the Import Flat File wizard

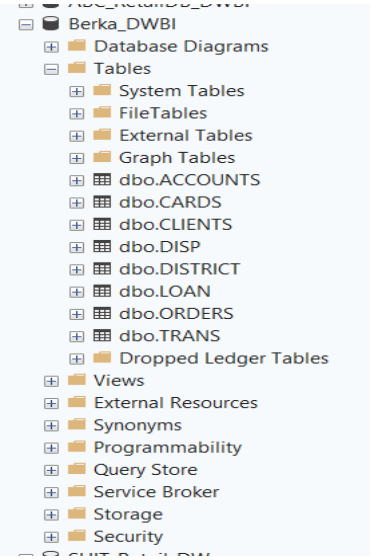


Figure 2.4: Source tables imported into the Berka_DWBI database

2.3.2 Checking record counts and column structures

	TableName	TotalRecords
1	ACCOUNTS	4500
2	CLIENTS	5369
3	DISP	5369
4	DISTRICT	77
5	CARDS	892
6	LOAN	682
7	ORDERS	6471
8	TRANS	1056320

Figure 2.5: Record counts of the imported source tables

The record counts of all eight source tables were validated after importing the .ASC files into SQL Server. The counts matched the original Berka dataset, confirming that the source data was imported completely.

2.3.3 Checking missing values in key identifier fields

	TableName	TotalRecords	Key_NotNull	Key_Null
1	ACCOUNTS	4500	4500	0
2	CLIENTS	5369	5369	0
3	DISP	5369	5369	0
4	DISTRICT	77	77	0
5	CARDS	892	892	0
6	LOAN	682	682	0
7	ORDERS	6471	6471	0
8	TRANS	1056320	1056320	0

Figure 2.6: Missing value check on key identifier fields

2.3.4 Checking duplicate values in key fields

	TableName	DuplicateGroups
1	ACCOUNTS	0
2	CLIENTS	0
3	DISP	0
4	DISTRICT	0
5	CARDS	0
6	LOAN	0
7	ORDERS	0
8	TRANS	0

Figure 2.7: Duplicate value check on key fields

2.3.5 Validating data types and date formats

	TableName	TotalRecords	InvalidDates
1	ACCOUNTS	4500	0
2	LOAN	682	0
3	CARDS	892	0
4	TRANS	1056320	0

Figure 2.8: Date format validation results

The data types of the source columns were reviewed using SQL Server metadata. The date fields in ACCOUNTS, LOAN, CARDS, and TRANS were validated against the YYMMDD format used by the source data. No invalid date values were identified.

2.3.6 Checking relationships between related tables

	Relationship	OrphanRecords
1	DISP ? CLIENTS	0
2	DISP ? ACCOUNTS	0
3	TRANS ? ACCOUNTS	0
4	ORDERS ? ACCOUNTS	0
5	LOAN ? ACCOUNTS	0
6	CARDS ? DISP	0

Figure 2.9: Referential integrity check between related tables

2.3.7 Identifying invalid or inconsistent values

Results Messages

	CheckField	InvalidRecords
1	ACCOUNTS - frequency	0
2	DISP - type	869
3	LOAN - status	0
4	CARDS - type	0
5	TRANS - type	16666
6	TRANS - operation	0
7	ORDERS - k_symbol	1379

Figure 2.10: Invalid and inconsistent value check

Categorical fields were profiled to identify unexpected or inconsistent values. The actual distinct values were compared with the source data description. Values such as DISPONENT in DISP and VYBER in TRANS were retained because they are present in the original source data. No categorical values were removed during source preparation.

2.3.8 Standardizing values where required

	Field	ValuesWithSpaces
1	ACCOUNTS.frequency	0
2	DISP.type	0
3	LOAN.status	0
4	CARDS.type	0
5	TRANS.type	0

Figure 2.11: Check for leading and trailing spaces in categorical fields

The main categorical fields were checked for unnecessary leading and trailing spaces. No additional text standardization was required for the checked fields. Date standardization will be performed during ETL by converting the source YYMMDD representation into the SQL DATE format.

2.3.9 Preparing clean data for the Data Warehouse

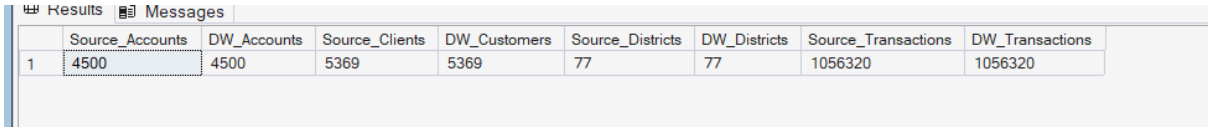
Based on the profiling and validation results, the source data was prepared for the ETL process. The source tables were retained without deleting valid source records. Required transformations, including date conversion and dimensional restructuring, will be performed during the ETL process.

2.4 Source Integration

The eight Berka .ASC source files were imported into Microsoft SQL Server and stored in the Berka_DWBI database as source tables. The source tables include account, client, disposition, district, loan, order, card, and transaction data. The relationships between the source tables were maintained using their identifier fields.

After importing the data, data profiling and quality validation were performed. Record counts were compared with the original dataset, key fields were checked for NULL and duplicate values, date formats were validated, and relationships were checked for orphan records. The validation results indicated that the imported source data was suitable for the next stage of the ETL process.

The prepared source tables will be used as the input layer for transforming and loading data into the dimensional Data Warehouse.



Results | Messages

	Source_Accounts	DW_Accounts	Source_Clients	DW_Customers	Source_Districts	DW_Districts	Source_Transactions	DW_Transactions
1	4500	4500	5369	5369	77	77	1056320	1056320

Figure 2.12: Source-to-Data Warehouse record count comparison

Task 3: Data Warehouse Architecture Design

3.1 Architecture Diagram

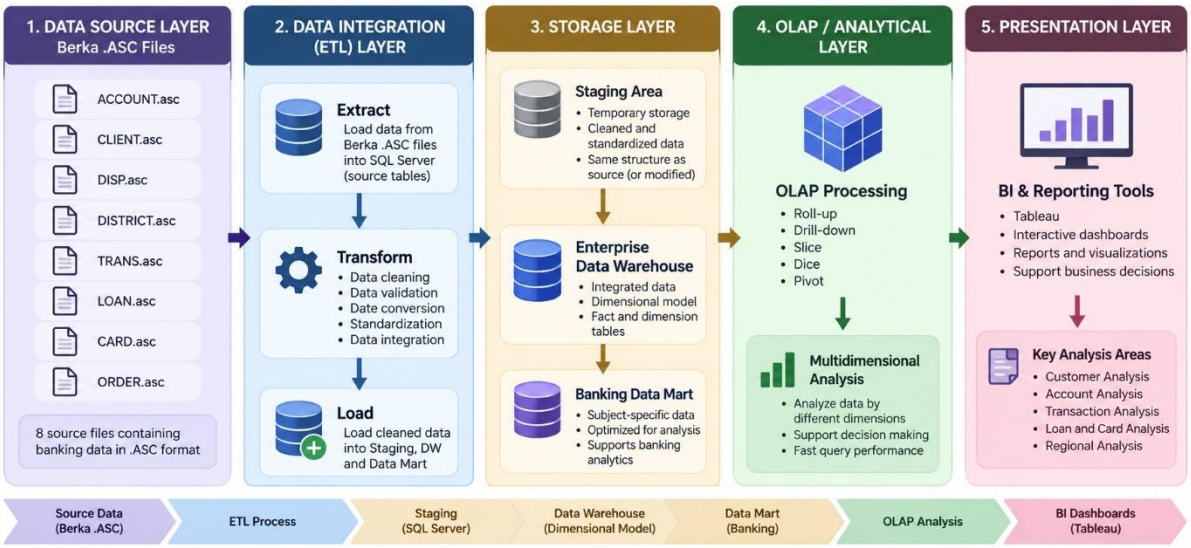


Figure 3.1: Data Warehouse Architecture for Berka Banking Data

3.2 Data Source Layer

The Data Source Layer contains the original Berka banking data provided as 6 related .ASC and 2 related .csv files. These files contain information about customers, accounts, transactions, loans, credit cards, payment orders, and district demographics. The files were imported into Microsoft SQL Server and stored as source tables in the Berka_DWBI database. The source relationships are maintained using common identifiers and provide the input for the subsequent ETL process.

3.3 Data Integration Layer

The Data Integration Layer is responsible for preparing, validating, and transferring the source data into the Data Warehouse. SQL scripts were used to perform source data validation and preprocessing, create the dimension and fact tables, and transfer the prepared data into the final Data Warehouse.

The data integration process involved checking source data quality, mapping source attributes to the corresponding dimension and fact table columns, and maintaining relationships

between the tables. The SQL scripts were saved separately to document the preprocessing, validation, and data transfer operations.

This process prepares the banking data for storage in the dimensional Data Warehouse and subsequent use in the Data Mart for analytical reporting.

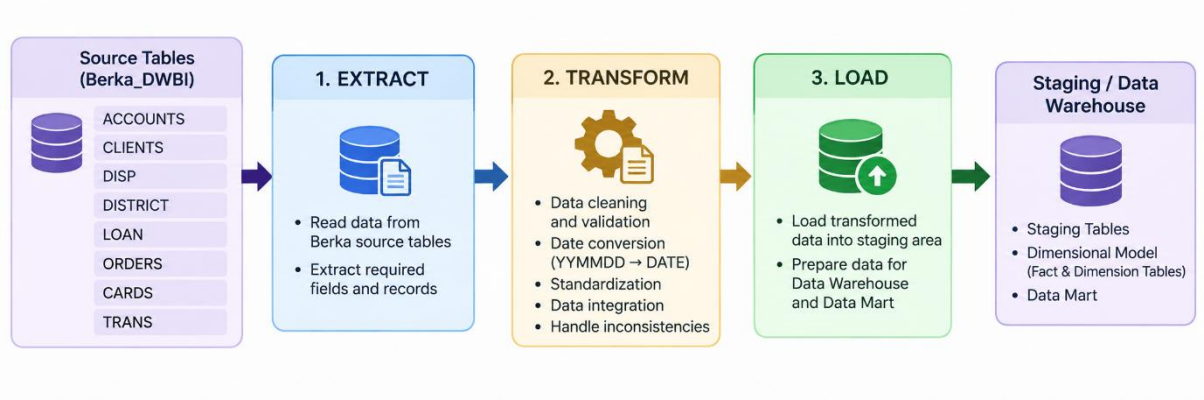


Figure 3.2: ETL process: extract, transform and load

3.4 Storage Layer

The Storage Layer stores the prepared banking data in a dimensional structure for analysis and reporting. The final Data Warehouse was created in the Berka_DW database, containing dimension tables and the fact table. The Berka_DataMart database was created to support analytical queries and business intelligence reporting.

The dimensional model organizes banking data into related tables, allowing transactions to be analyzed by customers, accounts, dates, districts, and transaction types.

3.4.1 Staging Area

The staging area is a temporary storage area used during the ETL process. Extracted data is prepared in the staging area before being loaded into the Data Warehouse.

Its main purposes are to:

- Store extracted data temporarily.
- Clean and validate the data.
- Standardize data formats.
- Prepare data for warehouse loading.
- Keep source data separate from the warehouse.

3.4.2 Enterprise Data Warehouse

The Enterprise Data Warehouse stores the integrated and structured banking data. It uses a dimensional model consisting of fact tables and dimension tables. The warehouse supports historical analysis of transactions, accounts, customers, and other banking information.

3.4.3 Data Mart

A Data Mart provides a focused view of the Data Warehouse for a specific analytical area. For this project, a Banking Transaction Data Mart will be used to support transaction analysis.

It can support the analysis of:

- Transaction amounts.
- Transaction types.
- Account activity.
- Customer activity.
- Time-based transaction trends.
-

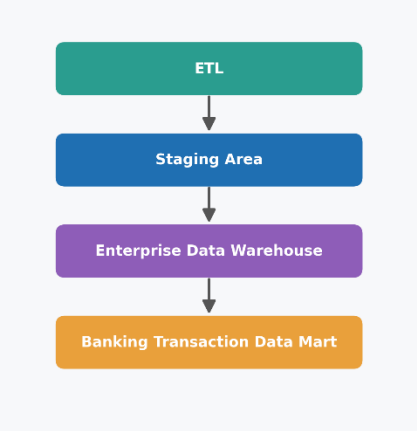


Figure 3.3: Flow of data through the storage layer

3.5 OLAP Layer

The OLAP Layer provides multidimensional analysis of the data stored in the Data Warehouse. It allows banking data to be analysed using dimensions such as **Time, Account, Customer, and Transaction Type**.

In this project, the OLAP operations are performed on the Berka_DW star schema and the Banking Transaction Data Mart. FactTransaction provides the measures (Amount, Balance, and TransactionCount), and the six dimension tables provide the analysis dimensions and hierarchies (Date: Year → Quarter → Month → Day; District: Region → District). The OLAP operations are demonstrated through the Power BI dashboard in Task 7.

The OLAP layer supports the following operations:

- **Roll-up:** Summarizes data from Day → Month → Year.
- **Drill-down:** Moves from Year → Month → Day for detailed analysis.
- **Slice:** Analyses data for a single selected value, such as one transaction type.
- **Dice:** Analyses a selected combination of values, such as specific years and transaction types.
- **Pivot:** Changes the dimensions used in the analytical view.

These operations allow users to analyse banking data at different levels of detail and from different perspectives.

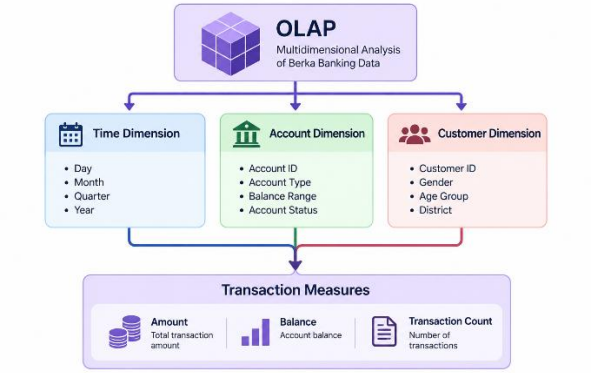


Figure 3.4: OLAP layer: dimensions and transaction measures

3.6 Presentation Layer

The Presentation Layer provides the interface through which users can view and analyse data warehouse information. It includes reports, dashboards, charts, and interactive visualizations.

In this project, Microsoft Power BI is used as the business intelligence visualization tool. The dashboard provides interactive views of banking transactions, transaction trends, customer activity, and transaction types.

Power BI uses the Berka_DW star schema and the Banking Transaction Data Mart as data sources. Filters and slicers allow users to explore the data based on selected dates, customers, operations, and transaction types.

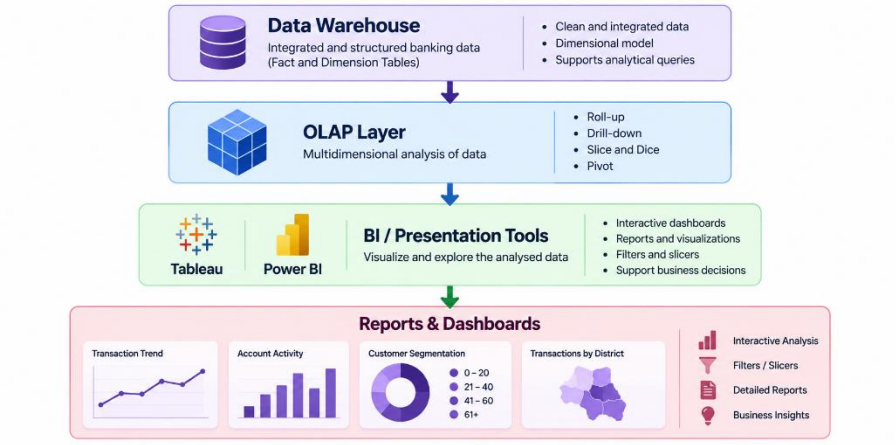


Figure 3.5: Presentation layer: BI tools, reports and dashboards

Task 4: Dimensional Data Warehouse Design and Implementation

4.1 Business Process

Banking Transaction Analysis

The selected business process for the Data Warehouse is Banking Transaction Analysis.

This process focuses on analysing banking transactions performed through customer accounts. It allows the bank to analyse transaction amounts, account balances, transaction types, operations, customers, districts, and transaction dates.

The Data Warehouse will support analysis such as:

- Total transaction amount over time.
- Transaction activity by account.
- Transaction activity by customer.
- Transaction activity by transaction type.
- Transaction activity across different districts.
- Changes in transaction activity over different time periods.

The TRANS source contains 1,056,320 transaction records with attributes such as transaction date, transaction type, operation, amount, balance, and account ID. Therefore, it provides the main source for the selected business process.

Why this business process?

The banking dataset contains a large number of transaction records and related customer, account, and district information. These related data can be integrated into a dimensional model to support historical analysis and BI reporting. This follows the dimensional modelling approach of first identifying the business process before defining the grain, dimensions, and fact table

4.2 Grain

Grain of FactTransaction

The grain of the FactTransaction table is defined as:

One row represents one individual banking transaction recorded in the TRANS source table.

The transaction table contains individual transaction records with attributes such as transaction ID, account ID, date, transaction type, operation, amount, and balance. Therefore, each transaction will be stored as a separate row in the fact table.

Why this grain is selected

This grain keeps the lowest level of transaction detail in the Data Warehouse. The lecture explains that fact tables should generally store data at the lowest possible level of detail so that different analyses and aggregations can be performed later.

Table 4.1: Example of the transaction-level grain

Transaction	Account	Date	Amount
T001	A001	1995-01-01	500
T002	A001	1995-01-02	250
T003	A002	1995-01-02	1000

Each transaction remains a separate fact record.

This allows the Data Warehouse to later calculate:

- Daily transaction amounts
- Monthly transaction amounts
- Transaction counts
- Account-level activity
- Customer-level activity
- Transaction activity by type and operation

The grain of the FactTransaction table is one row per individual banking transaction. No aggregation is performed at the fact-table loading stage, allowing detailed transaction-level analysis and later aggregation through OLAP operations.

4.3 Fact Table & Measures

Fact Table: FactTransaction

The central fact table in the Data Warehouse is **FactTransaction**. It stores the measurable information related to individual banking transactions.

The fact table will use **TransactionKey** as its primary key and will contain foreign keys that connect the transaction records with the relevant dimension tables.

Table 4.2: FactTransaction table structure

Column	Key / Type	Description
TransactionKey	PK	Unique warehouse key for each fact record
TransactionID	Source ID	Original transaction identifier from TRANS
DateKey	FK	Links to DimDate
AccountKey	FK	Links to DimAccount
CustomerKey	FK	Links to DimCustomer
DistrictKey	FK	Links to DimDistrict
TransactionTypeKey	FK	Links to DimTransactionType
OperationKey	FK	Links to DimOperation
Amount	Measure	Amount involved in the transaction
Balance	Measure	Account balance recorded for the transaction
TransactionCount	Measure	Represents one transaction

The source TRANS table contains the transaction ID, account ID, date, type, operation, amount, and balance used as the main inputs for this fact table.

Measures

The main measures are:

1. **Transaction Amount (Amount)**
   Stores the monetary amount associated with each transaction. It can be aggregated to analyse transaction amounts by date, account, customer, transaction type, or district.
2. **Account Balance (Balance)**
   Stores the account balance recorded for the transaction. It can be used to analyse balance patterns across different dimensions.
3. **Transaction Count (TransactionCount)**
   Stores a value of **1 for each fact record**. It can be summed to calculate the number of transactions.

Why these measures?

These measures represent numerical information associated with the banking transaction business process. The dimensional modelling lecture describes fact tables as storing measurable business information, while dimensions provide descriptive information.

The FactTransaction table stores one record for each banking transaction according to the defined grain. It contains foreign keys to the dimension tables and the measures required for transaction analysis. The main measures are transaction amount, account balance, and transaction count.

4.4 Dimension Tables

The dimension tables store the descriptive information used to analyse the measures in FactTransaction. The proposed Data Warehouse contains six dimensions: **Date, Account, Customer, District, Transaction Type, and Operation**.

4.4.1 DimDate

`DimDate` stores information about the date of each transaction.

Table 4.3: DimDate structure

Attribute	Description
DateKey (PK)	Unique warehouse key
FullDate	Complete transaction date
Day	Day number
Month	Month number
MonthName	Name of the month
Quarter	Quarter of the year
Year	Year

Hierarchy:

Year → Quarter → Month → Day

This supports time-based OLAP analysis such as yearly, monthly, and daily transaction analysis. The lecture and Lab 03 use similar time hierarchies for dimensional modelling.

4.4.2 DimAccount

DimAccount stores descriptive information about bank accounts.

Table 4.4: DimAccount structure

Attribute	Description
AccountKey (PK)	Unique warehouse key
AccountID	Original account identifier
Frequency	Account statement frequency
AccountStartDate	Date associated with the account

The source ACCOUNT relation contains account characteristics including account ID, district ID, frequency, and date.

4.4.3 DimCustomer

DimCustomer stores information about bank customers.

Table 4.5: DimCustomer structure

Attribute	Description
CustomerKey (PK)	Unique warehouse key
ClientID	Original customer identifier
BirthDate	Customer birth information
Gender	Customer gender derived from the source birth number
DistrictID	Customer's district identifier

The CLIENT source contains customer characteristics, while DISP provides the relationship between customers and accounts.

4.4.4 DimDistrict

DimDistrict stores geographical and demographic information related to districts.

Table 4.6: DimDistrict structure

Attribute	Description
DistrictKey (PK)	Unique warehouse key
DistrictID	Original district identifier
DistrictName	District name
Region	Geographic region
Population	Number of inhabitants
AverageSalary	Average salary
UnemploymentRate	Unemployment information

The original Berka data description provides district information including region, population, average salary, unemployment rates, and other demographic attributes.

Hierarchy:

Region → District

4.4.5 DimTransactionType

DimTransactionType stores the type of banking transaction.

Table 4.7: DimTransactionType structure

Attribute	Description
TransactionTypeKey (PK)	Unique warehouse key
TransactionType	Transaction type from the source

Examples of source values include:

- PRIJEM
- VYDAJ
- VYBER

These source values are retained as valid transaction categories.

4.4.6 DimOperation

DimOperation stores the operation associated with each banking transaction.

Table 4.8: DimOperation structure

Attribute	Description
OperationKey (PK)	Unique warehouse key
Operation	Transaction operation from the source

Examples include:

- VYBER
- PREVOD NA UCET
- VKLAD
- PREVOD Z UCTU
- VYBER KARTOU

The source TRANS table contains both transaction type and operation information.

The dimension tables provide descriptive information for analysing the measures stored in FactTransaction. The dimensions allow users to analyse banking transactions from different perspectives, including time, account, customer, district, transaction type, and operation.

4.5 Foreign Keys & Relationships

The FactTransaction table is connected to the dimension tables using foreign keys. These relationships allow the transaction measures to be analysed using different descriptive attributes.

Table 4.9: Foreign key relationships between FactTransaction and the dimension tables

Fact Table Foreign Key	Dimension Table	Relationship
DateKey	DimDate(DateKey)	Many transactions → One date
AccountKey	DimAccount(AccountKey)	Many transactions → One account
CustomerKey	DimCustomer(CustomerKey)	Many transactions → One customer
DistrictKey	DimDistrict(DistrictKey)	Many transactions → One district
TransactionTypeKey	DimTransactionType(TransactionTypeKey)	Many transactions → One transaction type
OperationKey	DimOperation(OperationKey)	Many transactions → One operation

The relationships follow the **star schema** structure, where the fact table is at the centre and the dimension tables are directly connected to it. In a star schema, dimension tables are not directly joined to each other.

Source-to-Warehouse Relationships

The dimensional model is based on the relationships identified in the Berka source data:

- TRANS.account_id is related to ACCOUNT.account_id.
- DISP connects clients and accounts through client_id and account_id.
- CARD is connected through disp_id.
- District information is connected using district_id.

During ETL, these source relationships will be used to identify the appropriate dimension records and obtain their surrogate keys before loading

The FactTransaction table uses foreign keys to connect transaction records with the Date, Account, Customer, District, Transaction Type, and Operation dimensions. These relationships allow the transaction measures to be analysed from multiple perspectives while maintaining the star schema structure.

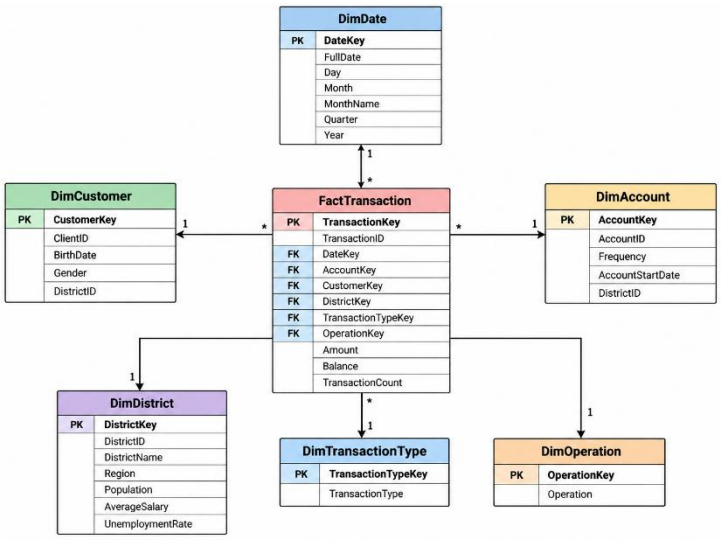


Figure 4.1: Foreign key relationships between FactTransaction and the dimension tables

4.6 Schema Selection

Selected Schema: Star Schema

A **Star Schema** is selected for the proposed Data Warehouse. In this schema, the FactTransaction table is placed at the centre and is directly connected to the surrounding dimension tables.

The selected dimensions are:

- DimDate
- DimAccount
- DimCustomer
- DimDistrict
- DimTransactionType
- DimOperation

The Star Schema is suitable for this project because the main business process is **banking transaction analysis**. The fact table stores the transaction measures, while the dimension tables provide descriptive information for analysis.

The dimensional modelling lecture describes a Star Schema as a structure where a fact table is connected directly to the surrounding dimension tables, without direct relationships between the dimension tables

Schema Structure

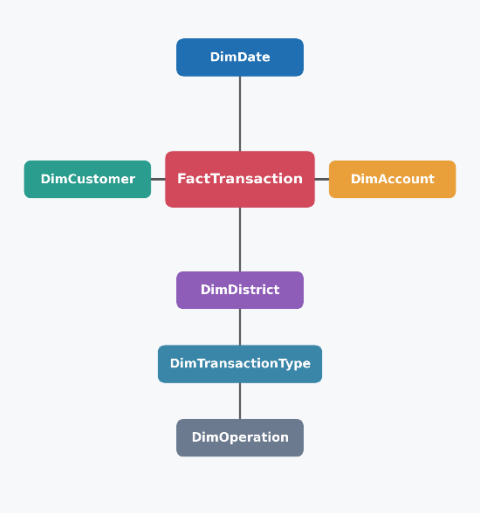


Figure 4.2: Star schema of the Berka Data Warehouse

Advantages for this project

- Simple structure for BI analysis.
- Easy to understand and maintain.
- Supports efficient analytical queries.
- Supports OLAP operations such as roll-up and drill-down.
- Suitable for connecting with Power BI.
- Keeps measurable transaction data in the fact table and descriptive information in dimensions.

A Star Schema is selected for the proposed Data Warehouse. The FactTransaction table is placed at the centre and is directly connected to the DimDate, DimAccount, DimCustomer, DimDistrict, DimTransactionType, and DimOperation tables. This structure separates measurable transaction data from descriptive information and provides a simple structure for analytical queries, OLAP operations, and BI reporting.

4.7 Dimension Hierarchies

Hierarchies allow users to move between different levels of detail during OLAP analysis. These hierarchies support operations such as **roll-up and drill-down**.

Date Hierarchy

This allows users to:

- Roll up daily transactions to monthly or yearly summaries.
- Drill down from yearly transactions to monthly and daily details.

District Hierarchy

A geographical hierarchy can be created using the district information:

This allows transaction activity to be analysed at the regional level and then at the district level.

The Berka source contains both **region** and **district-level** information.

Other Dimensions

The DimAccount, DimCustomer, DimTransactionType, and DimOperation dimensions do not have a clearly defined multi-level hierarchy based on the selected source attributes. Therefore, they will be used mainly for filtering and grouping rather than forcing an artificial hierarchy.

Table 4.10: Hierarchy summary of the dimensions

Dimension	Hierarchy
DimDate	Year → Quarter → Month → Day
DimDistrict	Region → District
DimAccount	No defined hierarchy
DimCustomer	No defined hierarchy
DimTransactionType	No defined hierarchy
DimOperation	No defined hierarchy

The Data Warehouse includes hierarchies mainly in the Date and District dimensions. The Date hierarchy supports time-based roll-up and drill-down from Day to Year, while the District hierarchy supports analysis from Region to District. Other dimensions will be used for filtering and grouping without introducing unsupported hierarchies. These hierarchies support roll-up and drill-down analysis in the Power BI reports.

4.8 Table Relationships

The Data Warehouse uses a **star schema**, where the FactTransaction table is connected directly to all dimension tables. The relationships are established using primary keys in the dimension tables and corresponding foreign keys in the fact table. This follows the star-schema structure described in the lecture, where dimension tables are connected to the central fact table and not directly to one another.

Table 4.11: Primary key to foreign key relationships

Dimension Table	Primary Key	Fact Table Foreign Key	Relationship
DimDate	DateKey	FactTransaction.DateKey	1 : Many
DimAccount	AccountKey	FactTransaction.AccountKey	1 : Many
DimCustomer	CustomerKey	FactTransaction.CustomerKey	1 : Many
DimDistrict	DistrictKey	FactTransaction.DistrictKey	1 : Many
DimTransactionType	TransactionTypeKey	FactTransaction.TransactionTypeKey	1 : Many
DimOperation	OperationKey	FactTransaction.OperationKey	1 : Many

Relationship Explanation

- **DimDate → FactTransaction:** One date can be associated with many transactions.
- **DimAccount → FactTransaction:** One account can have many transactions.
- **DimCustomer → FactTransaction:** One customer can be associated with many transaction records.
- **DimDistrict → FactTransaction:** One district can be associated with many transactions.
- **DimTransactionType → FactTransaction:** One transaction type can occur in many transactions.
- **DimOperation → FactTransaction:** One operation can occur in many transactions.

Therefore, the general relationship between each dimension and the fact table is:

One dimension record → Many fact records (1:M)

Schema Relationship

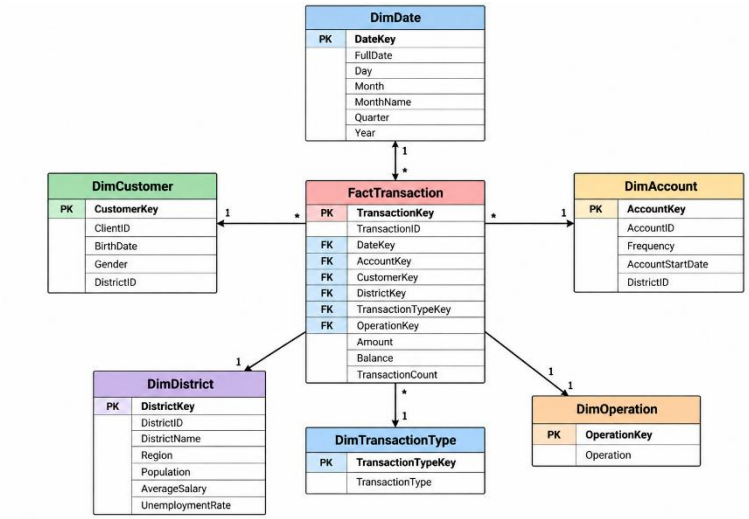


Figure 4.3: Table relationships between the dimension tables and FactTransaction

The FactTransaction table is related to each dimension table through foreign keys. Each dimension has a one-to-many relationship with the fact table because one dimension record can be associated with multiple transaction records. The dimension tables are not directly connected to each other. This structure maintains the Star Schema and supports efficient multidimensional analysis.

4.9 Table Descriptions

The proposed Data Warehouse consists of one fact table and six dimension tables. The fact table stores transaction-level measures, while the dimension tables store descriptive information used for analysis.

4.9.1 FactTransaction

Table 4.12: FactTransaction attribute descriptions

Attribute	Description
TransactionKey	Surrogate primary key that uniquely identifies each fact record.
TransactionID	Original transaction identifier from the TRANS source table.
DateKey	Foreign key linking the transaction to DimDate.
AccountKey	Foreign key linking the transaction to DimAccount.
CustomerKey	Foreign key linking the transaction to DimCustomer.
DistrictKey	Foreign key linking the transaction to DimDistrict.
TransactionTypeKey	Foreign key linking the transaction to DimTransactionType.
OperationKey	Foreign key linking the transaction to DimOperation.
Amount	Monetary amount associated with the transaction.
Balance	Account balance recorded for the transaction.
TransactionCount	Measure representing one transaction record.

The TRANS source records one transaction on an account and contains attributes such as transaction ID, account ID, date, type, operation, amount, and balance.

4.9.2 DimDate

DimDate provides calendar information for analysing transactions over time.

Table 4.13: DimDate attribute descriptions

Attribute	Description
DateKey	Primary key of the date dimension.
FullDate	Complete transaction date.
Day	Day component of the date.
Month	Month component of the date.
MonthName	Name of the month.
Quarter	Quarter of the year.
Year	Year of the transaction.

Hierarchy: Year → Quarter → Month → Day

4.9.3 DimAccount

DimAccount stores descriptive information about bank accounts.

Table 4.14: DimAccount attribute descriptions

Attribute	Description
AccountKey	Primary key of the account dimension.
AccountID	Original account identifier.
Frequency	Frequency of account statement issuance.
AccountStartDate	Date on which the account was created.

The source ACCOUNT relation contains the account identifier, branch district, creation date, and statement frequency.

4.9.4 DimCustomer

DimCustomer stores descriptive information about customers.

Table 4.15: DimCustomer attribute descriptions

Attribute	Description
CustomerKey	Primary key of the customer dimension.
ClientID	Original customer identifier.
BirthDate	Customer date of birth derived from the source birth number.
Gender	Customer gender derived from the source birth number.
DistrictID	District identifier associated with the customer's address.

The source documentation states that the client's **birth number contains the birthday and sex**, while district_id represents the client's address. Therefore, BirthDate and Gender can be derived during ETL.

4.9.5 DimDistrict

DimDistrict stores geographical and demographic information about the district associated with the bank account branch.

Table 4.16: DimDistrict attribute descriptions

Attribute	Description
DistrictKey	Primary key of the district dimension.
DistrictID	Original district identifier.
DistrictName	Name of the district.
Region	Region containing the district.
Population	Number of inhabitants.
AverageSalary	Average salary in the district.
UnemploymentRate	District unemployment information.

The source documentation identifies district_id, district name, region, population, average salary, and unemployment rates as demographic attributes.

Hierarchy: Region → District

Design clarification: In this model, DimDistrict represents the district associated with the bank account (ACCOUNTS.district_id), whereas DimCustomer.DistrictID stores the district of the customer’s address and is treated separately.

4.10 Design Assumptions

The following assumptions were made when designing and implementing the Data Warehouse:

1. **Business Process**
   The main business process is **Banking Transaction Analysis**.
2. **Fact Table Grain**
   Each row in FactTransaction represents **one individual banking transaction** from the TRANS source table.
3. **Customer Mapping**
   A transaction is associated with the customer who is identified as the **OWNER** of the related account in the DISP table.
4. **Account District**
   DimDistrict represents the district associated with the bank account through ACCOUNTS.district_id.
5. **Customer District**
   DimCustomer.DistrictID represents the district associated with the customer's address. It is treated separately from the account district.
6. **Surrogate Keys**
   Surrogate keys are used as primary keys in the dimension tables to provide stable warehouse identifiers.
7. **Source Identifiers**
   Original source identifiers such as `ClientID`, `AccountID`, and `TransactionID` are retained for traceability.
8. **Date Conversion**
   Dates stored in the source `YYMMDD` format are converted into SQL Server `DATE` values during the ETL process.
9. **Transaction Amount**
   `Amount` is treated as an **additive measure** and can be aggregated across the dimensions.
10. **Account Balance**
    `Balance` is treated as a **semi-additive measure**. It can be analysed across dimensions such as account and customer, but summing balances across time is not considered meaningful.
11. **Transaction Count**
    `TransactionCount` is assigned a value of `1` for each fact record. Therefore, it can be summed to calculate the number of transactions.
12. **Missing Operations**
    Transactions with a missing operation are not removed. They are assigned to the **`Unknown`** member in `DimOperation` to preserve the transaction-level grain.
13. **Star Schema**
    The Data Warehouse uses a star schema where `FactTransaction` is directly connected to the six dimension tables. Dimension tables are not directly connected to each other.

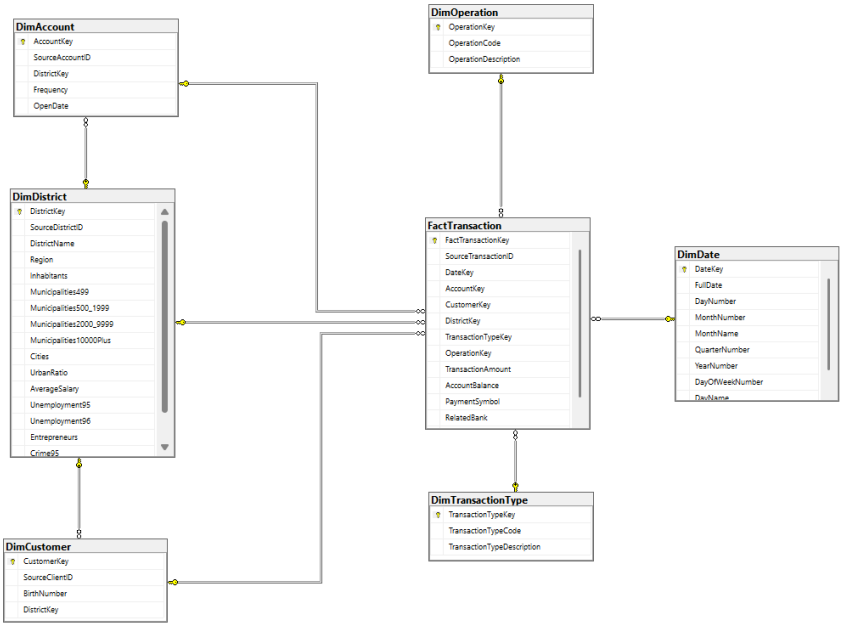


Figure 4.4: Database diagram of the implemented star schema in SQL Server

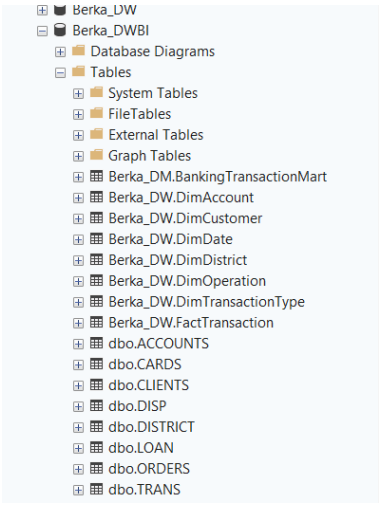


Figure 4.5: Data Warehouse tables in SQL Server Object Explorer

Task 5: ETL Process Development

5.1 ETL Overview

The ETL process was developed to extract data from the Berka source tables, transform and validate the data, and load the processed data into the **Berka Data Warehouse**. The process integrates customer, account, district, and transaction information into a dimensional structure for analysis.

The ETL process consists of three main stages:

Extract → Transform → Load

ETL Workflow

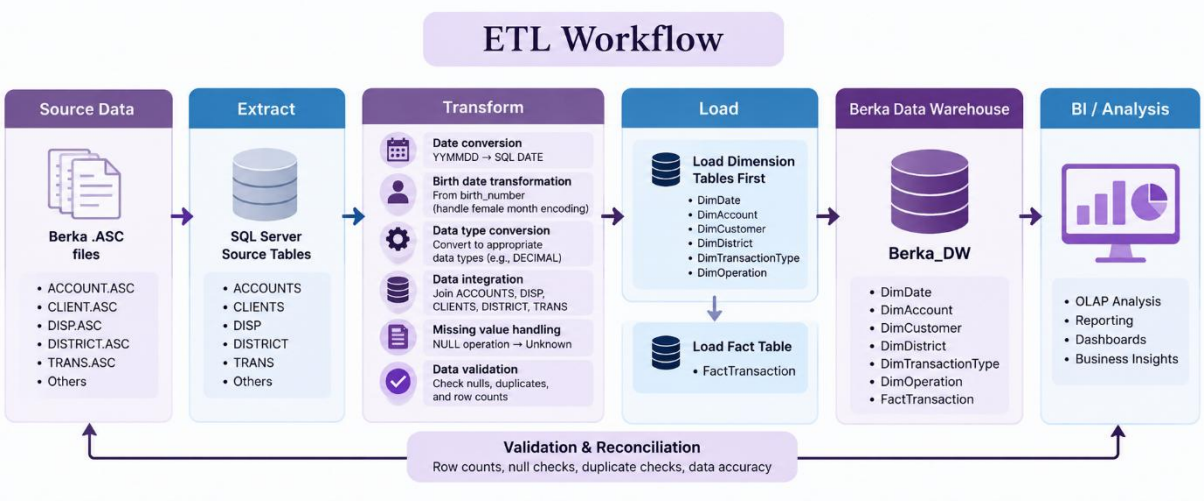


Figure 5.1: ETL workflow from the source files to the Data Warehouse

During the **Extract** stage, the Berka .ASC files were imported into SQL Server as source tables. During the **Transform** stage, the data was cleaned, converted, integrated, and validated before loading. During the **Load** stage, the transformed data was inserted into the dimension tables and the FactTransaction table.

The ETL process also preserves the transaction-level grain by retaining all **1,056,320 transaction records**. Missing transaction operations were assigned to an Unknown category instead of removing the records.

This ETL process provides the required prepared and integrated data for the **Berka Data Warehouse and subsequent BI analysis**.

5.1.1 ETL Implementation Using SQL Server

The ETL process was implemented in Microsoft SQL Server using T-SQL scripts. The extraction, transformation, and loading logic is organised as a sequence of SQL scripts that read from the Berka_DWBI source tables and write into the Berka_DW tables. The dimension tables are loaded before FactTransaction so that the surrogate keys are available during the fact load.

Table 5.1: Mapping of ETL operations to T-SQL techniques

ETL Operation	T-SQL Technique	Purpose
Extract source data	SELECT from source tables	Reads the source tables from the Berka_DWBI database.
Date conversion	CONVERT / TRY_CONVERT	Converts YYMMDD values to the SQL Server DATE type.
Birth number transformation	SUBSTRING and CASE	Derives BirthDate and Gender from the birth_number field.
Data type conversion	CAST / TRY_CONVERT	Converts Amount, Balance, and district measures to the required warehouse data types.
Missing operation handling	ISNULL / COALESCE	Assigns missing operation values to the Unknown member of DimOperation.
Surrogate key mapping	JOIN to dimension tables	Retrieves the dimension surrogate keys while loading FactTransaction.
Load dimension and fact tables	INSERT INTO ... SELECT	Inserts the transformed rows into the Data Warehouse tables.
Execution order	Ordered script execution	Ensures that the dimension loads complete before the fact load.

5.2 Extract

The extraction stage collects data from the Berka dataset and makes it available in SQL Server for further ETL processing. The Berka dataset was provided as multiple .ASC files representing related banking entities.

The source files were imported into Microsoft SQL Server and stored as source tables. The main source tables used in the Data Warehouse implementation were:

Table 5.2: Source files and their SQL Server tables

Source File	SQL Server Table	Purpose
ACCOUNT.ASC	ACCOUNTS	Account information
CLIENT.ASC	CLIENTS	Customer information
DISP.ASC	DISP	Customer–account relationships
DISTRICT.ASC	DISTRICT	District information
TRANS.CSV	TRANS	Banking transactions
ORDER.CSV	ORDERS	Payment orders
LOAN.ASC	LOAN	Loan information
CARD.ASC	CARDS	Card information

The .ASC files were first imported into SQL Server. The imported tables were then used as the source layer for the ETL process. The source dataset contains **1,056,320 transaction records**, making TRANS the main source for the transaction fact table.

The extracted source data was checked before transformation. Key checks included record counts, null values, duplicate identifiers, data types, date values, and relationships between related source tables.

5.3 Transform

The transformation stage prepares the extracted source data for loading into the **Berka Data Warehouse**. The data was cleaned, converted, integrated, and validated before loading into the dimension and fact tables.

5.3.1 Date Conversion

The Berka source stores dates in YYMMDD format. These values were converted into SQL Server DATE values during the ETL process.

For example:

950324 → 1995-03-24

This transformation was applied when loading DimAccount, DimCustomer, and FactTransaction.

5.3.2 Customer Birth Date Transformation

The birth_number field in the CLIENTS table was transformed into a proper birth date.

The transformation also handles the female month encoding used in the source data. The resulting values were stored in DimCustomer.BirthDate.

A validation check confirmed that:

FutureBirthDates = 0

Therefore, no future birth dates were loaded into the dimension.

5.3.3 Data Type Conversion

Source values were converted to suitable warehouse data types.

Examples include:

- Amount → DECIMAL(18,2)
- Balance → DECIMAL(18,2)
- AverageSalary → DECIMAL(10,2)
- UnemploymentRate → DECIMAL(10,2)
- Source date values → DATE
-

5.3.4 Missing Value Handling

The operation field in the TRANS table contains missing values. Instead of removing these transactions, the missing values were assigned to an **Unknown** member in DimOperation.

The validation showed:

UnknownOperationTransactions = 183,114

This allowed all transaction records to remain in the fact table.

5.3.5 Data Integration

Related source tables were joined during the transformation process.

The main relationship used for customer identification was:

TRANS –(account_id)→ ACCOUNTS –(account_id)→ DISP –(client_id)→ CLIENTS

Only the OWNER relationship in DISP was used to associate an account transaction with its customer.

The validation showed:

TransactionsWithoutOwner = 0
AccountsWithoutOwner = 0

District Data Transformation

District information was transformed before loading into DimDistrict. A non-numeric ? value in the unemployment field was handled using TRY_CONVERT() so that it became NULL instead of causing the load to fail.

The result was:

Districts loaded = 77
NULL unemployment values = 1

5.3.6 Data Validation

Several validation checks were performed after transformation and loading:

- Source and warehouse row counts
- Null foreign keys
- Duplicate transaction IDs
- Missing account owners
- Missing customer mappings
- Future birth dates

The final fact table contained:

Source transactions = 1,056,320
Warehouse transactions = 1,056,320
Dropped transactions = 0

Therefore, the transformation process preserved the required transaction-level grain.

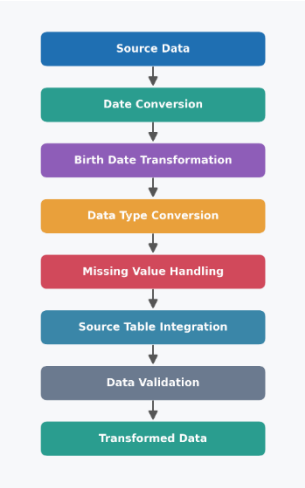


Figure 5.2: ETL transformation steps

5.4 Load

The load stage transfers the transformed data into the **Berka Data Warehouse**. The dimension tables were loaded first so that their surrogate keys were available when loading the fact table.

5.4.1 Loading Dimension Tables

The following dimension tables were populated:

- DimDate
- DimAccount
- DimCustomer
- DimDistrict
- DimTransactionType
- DimOperation

The source data was transformed during the loading process and inserted into the corresponding dimension tables.

The loaded row counts were:

Table 5.3: Rows loaded into each dimension table

Dimension	Rows Loaded
DimDate	2,191
DimAccount	4,500
DimCustomer	5,369
DimDistrict	77
DimTransactionType	3
DimOperation	6

5.4.2 Loading Fact Table

After loading the dimensions, the FactTransaction table was populated from the TRANS source.

During the fact loading process:

- DateKey was obtained from DimDate.
- AccountKey was obtained from DimAccount.
- CustomerKey was obtained using the account owner relationship.
- DistrictKey was obtained from DimDistrict.
- TransactionTypeKey was obtained from DimTransactionType.
- OperationKey was obtained from DimOperation.
- Amount and Balance were stored as measures.
- TransactionCount was assigned a value of 1 for each transaction.

5.4.3 Surrogate Key Mapping

Surrogate keys were used to connect the fact table with the dimension tables.

TRANS.account_id → DimAccount.AccountKey → FactTransaction.AccountKey

This approach allows the fact table to use warehouse keys rather than relying only on the original source identifiers.

5.4.4 Load Validation

The final validation confirmed that all transaction records were successfully loaded.

5.5 ETL Validation and Results

After the ETL process was completed, validation checks were performed to confirm that the transformed data was loaded correctly into the Data Warehouse.

5.5.1 Source-to-Warehouse Row Count Validation

The source and warehouse row counts were compared to identify any data loss during the loading process.

Table 5.4: Source and warehouse row count comparison

Table	Source Rows	Warehouse Rows
Accounts	4,500	4,500
Customers	5,369	5,369
Districts	77	77
Transactions	1,056,320	1,056,320

The results show that the expected records were successfully loaded into the corresponding warehouse tables.

5.5.2 Foreign Key Validation

The fact table was checked for NULL foreign keys.

Table 5.5: NULL foreign key check on FactTransaction

Foreign Key	NULL Records
DateKey	0
AccountKey	0
CustomerKey	0
DistrictKey	0
TransactionTypeKey	0
OperationKey	0

This confirms that every fact record has valid references to the required dimension records.

5.5.3 Duplicate Transaction Validation

The TransactionID values in FactTransaction were checked for duplicates.

Result: No duplicate transaction IDs were found.

This confirms that each source transaction was loaded as a separate fact record without unintended duplication.

5.5.4 Transaction Completeness

The final fact-table validation showed:

Source Transactions : 1,056,320
Loaded Transactions : 1,056,320
Dropped Transactions: 0

Therefore, all source transactions were retained during the ETL process.

5.5.5 Missing Operation Validation

The source contained transactions with missing operation values. These records were not removed. Instead, they were assigned to the Unknown member in DimOperation.

Unknown Operation Transactions : 183,114

This ensures that missing source values do not cause transaction records to be lost.

5.5.6 ETL Validation Summary

The validation results confirm that:

- Source and warehouse record counts match.
- No NULL foreign keys exist in the fact table.
- No duplicate transaction IDs were detected.
- All **1,056,320** transactions were retained.
- Missing operation values were handled using the Unknown category.
- The dimensional relationships were successfully established.

Therefore, the ETL process successfully transformed and loaded the Berka banking data into the **Berka Data Warehouse**.

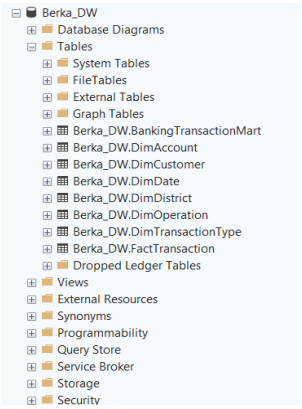


Figure 5.3: Data Warehouse and Data Mart tables in the Berka_DW database

Task 6: Data Mart Development

6.1 Data Mart Overview

A **Banking Transaction Data Mart** was designed to support focused analysis of banking transactions. It is derived from the validated Berka_DW Data Warehouse and provides data required for OLAP analysis and BI reporting.

The Data Mart focuses on transaction-related analysis, including transaction amounts, transaction counts, account activity, customer activity, transaction types, operations, time periods, and districts.

The main fact and dimension tables used are:

- FactTransaction
- DimDate
- DimAccount
- DimCustomer
- DimDistrict
- DimTransactionType
- DimOperation

The Data Mart provides a subject-specific analytical layer between the Data Warehouse and the OLAP/BI tools. This layer serves as the basis for OLAP analysis and the BI dashboards.

6.2 Data Mart Implementation

A Banking Transaction Data Mart was implemented from the validated Berka_DW Data Warehouse. The Data Mart was created under the Berka_DM schema using a table named BankingTransactionMart.

The Data Mart combines the transaction fact data with the relevant descriptive attributes from the Date, Account, Customer, District, Transaction Type, and Operation dimensions. This provides a simplified dataset for banking transaction analysis and BI reporting.

The Data Mart retains the transaction-level grain of the Data Warehouse, where each row represents one banking transaction. The following information is included:

- Transaction and transaction date information
- Account information
- Customer information
- District and regional information
- Transaction type
- Transaction operation
- Transaction amount
- Account balance
- Transaction count

The Data Mart was created using the following structure:

Berka_DW → Berka_DM → Power BI Dashboard (OLAP analysis)

Data Mart Validation

The implemented Data Mart contains **1,056,320 records**. This was compared with the FactTransaction table in the Data Warehouse.

Results | Messages

	DWRows	DataMartRows
1	1056320	1056320

Figure 6.1: Validation of Data Warehouse and Data Mart record counts

The results confirm that all transaction records from the Data Warehouse were retained when creating the Data Mart. Sample records were also inspected to verify that the date, account, customer, district, transaction type, operation, and measure information were correctly combined.

The following results were produced from the Data Mart: sample records, a summary by transaction type, and the monthly transaction trend.

Results | Messages

	TransactionID	TransactionDate	AccountID	CustomerID	TransactionType	Operation	TransactionAmount	AccountBalance
1	207264	1993-01-01	704	844	Credit	Credit in cash	1000.00	1000.00
2	171812	1993-01-01	576	692	Credit	Credit in cash	900.00	900.00
3	695247	1993-01-01	2378	2873	Credit	Credit in cash	700.00	700.00
4	1117247	1993-01-01	3818	4601	Credit	Credit in cash	600.00	600.00
5	579373	1993-01-02	1972	2397	Credit	Credit in cash	400.00	400.00
6	771035	1993-01-02	2632	3177	Credit	Credit in cash	1100.00	1100.00
7	452728	1993-01-03	1539	1866	Credit	Credit in cash	600.00	600.00
8	232960	1993-01-03	793	950	Credit	Credit in cash	800.00	800.00
9	505240	1993-01-03	1726	2093	Credit	Credit in cash	1000.00	1000.00
10	497211	1993-01-03	1695	2058	Credit	Credit in cash	200.00	200.00

Figure 6.2: Sample records from the Banking Transaction Data Mart

Results | Messages

	TransactionType	TotalTransactions	TotalAmount	AverageAmount
1	Credit	405083	3227472424.50	7967.434882
2	Withdrawal	634571	2821717377.80	4446.653530
3	Withdrawal in Cash	16666	208603758.00	12516.726149

Figure 6.3: Transaction count, total amount and average amount by transaction type

Results | Messages

	TransactionYear	TransactionMonth	TotalTransactions	TotalAmount
1	1993	1	177	702157.60
2	1993	2	395	2726925.30
3	1993	3	676	4730318.60
4	1993	4	913	7378367.80
5	1993	5	1306	11680753.40
6	1993	6	1880	18291948.90
7	1993	7	2399	18021294.70
8	1993	8	2938	21028811.60
9	1993	9	3441	23438701.60
10	1993	10	3989	27570899.70
11	1993	11	4422	28525741.10
12	1993	12	5669	40212608.40
13	1994	1	9139	42592232.50
14	1994	2	5601	31281776.70
15	1994	3	6255	37383802.60
16	1994	4	6549	37491620.60
17	1994	5	7059	41734136.90
18	1994	6	7916	59676368.50
19	1994	7	7398	42474508.70
20	1994	8	7600	43566089.10
21	1994	9	7844	46479839.60
22	1994	10	8197	46615494.00
23	1994	11	8275	47591232.60
24	1994	12	9795	61903629.00
25	1995	1	15493	66245579.70
26	1995	2	8682	45126076.70
27	1995	3	9304	54113431.30
28	1995	4	9519	54844143.60
29	1995	5	9938	59914969.20
30	1995	6	10930	81612628.10
31	1995	7	10431	59959795.50
32	1995	8	10766	61914115.80
33	1995	9	11013	63051038.80
34	1995	10	11431	66593805.40

Figure 6.4: Monthly transaction count and total amount in the Data Mart

The Banking Transaction Data Mart was successfully implemented and validated. It contains the same 1,056,320 transaction records as the Data Warehouse fact table, confirming that no transaction records were lost during Data Mart creation. The resulting focused dataset can be used for OLAP analysis and BI dashboard development.

6.3 Purpose, Target Users and Analytical Benefits

6.3.1 Purpose of the Data Mart

The purpose of the Banking Transaction Data Mart is to provide a focused dataset for analysing banking transaction activities. It simplifies access to the relevant transaction, account, customer, time, and district information stored in the Data Warehouse.

The Data Mart supports historical analysis of transaction amounts, transaction counts, transaction types, account activity, customer activity, and regional transaction patterns.

6.3.2 Target Users

The main target users of the Data Mart are:

- **Bank Management** – to monitor overall transaction activity and identify business trends.
- **Branch Managers** – to analyse transaction activity across accounts and districts.
- **Business Analysts** – to perform detailed transaction and customer analysis.
- **BI Analysts** – to create OLAP reports and interactive BI dashboards.

6.3.3 Analytical Benefits

The Banking Transaction Data Mart provides the following benefits:

- **Faster analysis** by providing a focused dataset for transaction-related queries.
- **Simplified reporting** because relevant data is available in a single analytical structure.
- **Time-based analysis** of transactions by day, month, quarter, and year.
- **Transaction analysis** by transaction type and operation.
- **Account and customer analysis** to understand transaction activity.
- **Regional analysis** using district and region information.
- **Improved BI reporting** by providing a suitable data source for OLAP analysis and dashboards.

The Data Mart therefore provides a focused analytical layer between the Data Warehouse and the BI reporting environment. This follows the lecture definition of a Data Mart as a smaller, department-focused subset of a Data Warehouse used for faster and simpler analysis.

Task 7: OLAP Analysis and Business Intelligence Dashboard Development

7.1 Introduction

The objective of this task was to analyze banking transaction data using Online Analytical Processing (OLAP) concepts and develop an interactive Business Intelligence (BI) dashboard. The Berka banking dataset was used to analyze transaction amounts, transaction volumes, customer demographics, regional performance, and transaction types. Power BI was used to visualize the data and support the analysis of banking activities.

7.2 OLAP Operations

OLAP operations allow users to analyze data from different perspectives and levels of detail.

7.2.1 Roll-up

Roll-up summarizes detailed data into higher-level information. In this project, transaction data can be summarized from monthly transaction details into yearly transaction totals to identify changes in banking activity over time.

7.2.2 Drill-down

Drill-down allows users to move from summarized information to more detailed information. For example, yearly transaction data can be explored at monthly or daily levels when the appropriate date hierarchy is available.

7.2.3 Slice

Slice selects a specific value from one dimension to analyze a subset of the data. For example, users can select a particular transaction type to examine its transaction amounts or volumes.

7.2.4 Dice

Dice filters data using multiple conditions across different dimensions. For example, transaction data can be analyzed for selected years, transaction types, and districts.

7.2.5 Pivot

Pivot changes the orientation of the data to provide a different analytical perspective. For example, transaction amounts can be compared across years and transaction types.

7.3 Power BI Dashboard Development

Power BI was connected to the Berka banking database, and the relevant fact and dimension tables were used to create visualizations. The dashboard was designed to present important transaction trends and comparisons in an understandable format.

The Trend Analysis page contains the following visualizations:

1. **Yearly Transaction Amount and Volume:** A combination chart displaying yearly transaction amounts by transaction type and the number of transactions.
2. **Monthly Transaction Volume:** A bar chart showing transaction counts by month.
3. **Customer Gender Comparison:** A pie chart comparing transaction counts associated with male and female customers.
4. **Yearly Regional Comparison:** A clustered column chart comparing transaction amounts across regions and years.
5. **Transaction Type by Year:** A stacked column chart comparing transaction volumes for different transaction types over time.
6. **Yearly Transaction Amount Comparison:** A doughnut chart showing the distribution of total transaction amounts across years.
7. **Transactions by District:** A bar chart comparing transaction counts across districts.
8. **Transaction Amount by Operation:** A doughnut chart showing the distribution of transaction amounts by banking operation.
9. **Average Transaction Amount by Type:** A column chart comparing transaction amounts across transaction types. The Amount aggregation should be set to Average for this analysis.

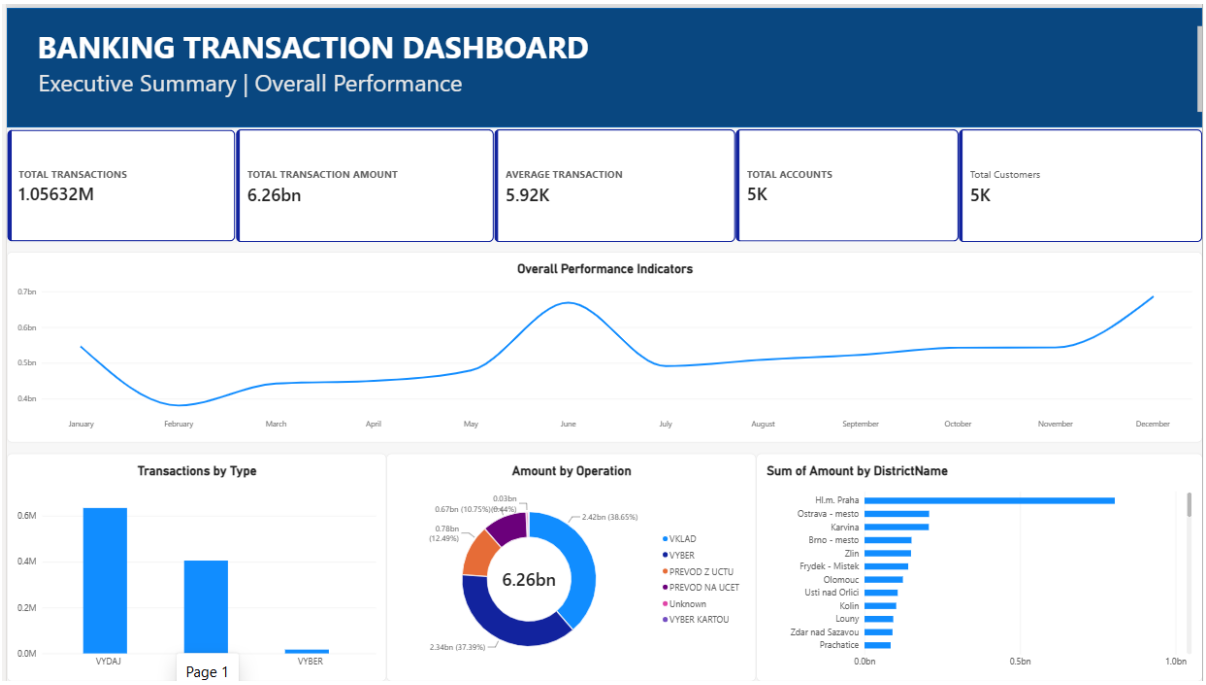


Figure 7.1: Executive Summary page of the Banking Transaction Dashboard



Figure 7.2: Trend Analysis page of the Banking Transaction Dashboard

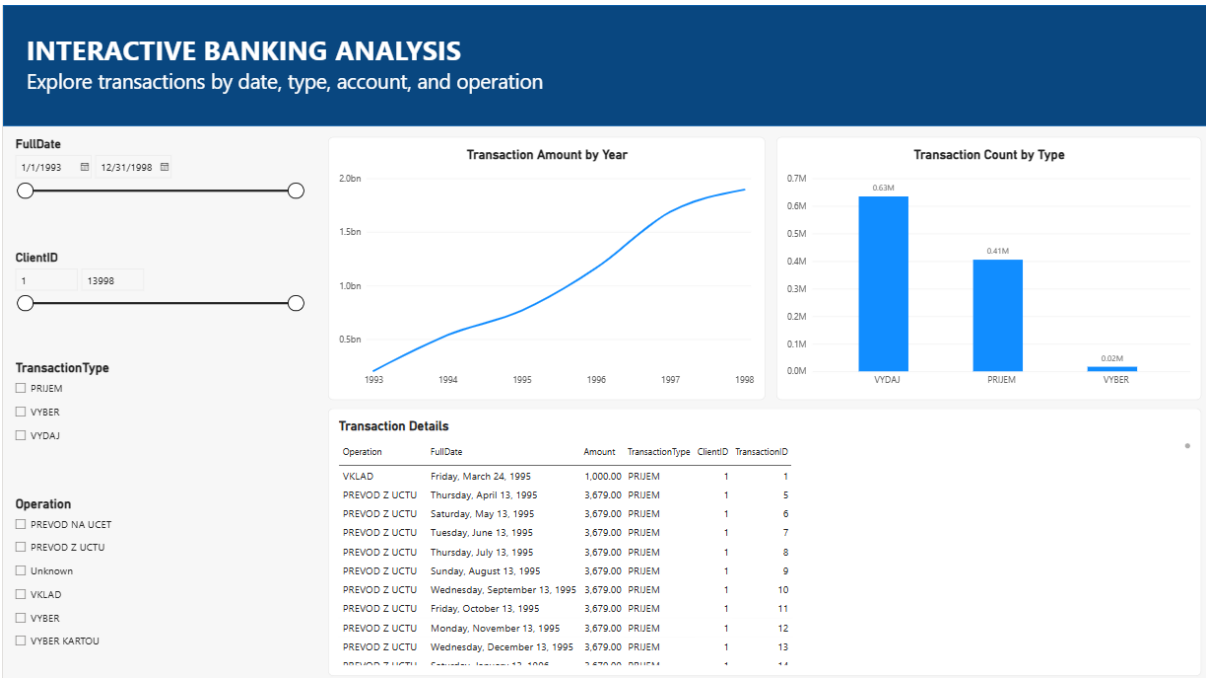


Figure 7.3: Interactive Banking Analysis page of the Banking Transaction Dashboard

7.4 Dashboard Analysis

The dashboard provides several perspectives on banking transaction activity. The yearly transaction amount and volume chart helps identify changes in transaction activity over time. Monthly transaction volume provides a view of transaction frequency by month.

The regional and district comparisons help examine geographical differences in transaction activity. The transaction type charts show how transaction categories contribute to transaction

volume and amount. The gender comparison provides a demographic perspective based on the transactions associated with each customer group.

These visualizations help users explore patterns, compare categories, and understand the distribution of banking transactions.

7.5 Conclusion

This task demonstrated the application of OLAP concepts and Business Intelligence visualization techniques to the Berka banking dataset. Power BI was used to create a dashboard containing time-based analysis, geographical comparisons, transaction type analysis, and customer demographic comparisons. The dashboard provides a visual summary of banking activities and supports further analysis of transaction patterns.

Task 8: Business Insights and Recommendations

8.1 Introduction

The Power BI dashboard developed using the Berka banking dataset provides insights into transaction trends, customer demographics, regional performance, transaction types, and banking operations. The following insights are derived from the visualizations and can support banking management in understanding transaction patterns and improving operational planning.

8.2 Business Insights

8.2.1 Yearly Transaction Amount Growth

The yearly transaction amount comparison shows an overall increase in transaction amounts from 1993 to 1998. The highest yearly amount is recorded in 1998, contributing approximately 30.28% of the total amount displayed in the chart.

Recommendation: Banking management can examine the factors associated with this growth and use historical transaction trends when planning future banking services and resource requirements.

8.2.2 Monthly Transaction Volume

The Monthly Transaction Volume chart shows differences in transaction counts across months. January has the highest displayed transaction volume, while several other months show lower volumes.

Recommendation: The bank can use monthly transaction patterns to plan staff availability, system capacity, and customer service resources. The analysis should consider seasonal effects before making long-term decisions.

8.2.3 Customer Gender Comparison

The gender comparison chart shows approximately 535.49K transactions associated with male customers (50.69%) and 520.83K transactions associated with female customers (49.31%). The distribution is relatively balanced.

Recommendation: The bank can maintain services that meet the needs of both customer groups. Further analysis of transaction amounts, account types, and service usage may help identify differences in customer behaviour.

8.2.4 Regional Transaction Differences

The Yearly Regional Comparison chart shows differences in transaction amounts between regions. Transaction amounts increase across the displayed years, and some regions contribute higher amounts than others.

Recommendation: Management can investigate regional differences to identify areas with greater banking activity and determine whether additional services, staff, or infrastructure are required.

8.2.5 Transaction Type Patterns

The Transaction Type by Year chart shows that transaction volumes vary by transaction type. The displayed transaction categories include PRIJEM, VYBER, and VYDAJ, with VYDAJ contributing a substantial portion of the transaction count in the later years.

Recommendation: The bank can monitor transaction types to understand customer banking activity and ensure that systems and operational processes can handle transaction demand efficiently.

8.2.6 District-Level Transaction Activity

The Transactions by District chart shows differences in transaction counts across districts. Hlm. Praha has the highest displayed transaction count, while other districts show lower volumes.

Recommendation: The bank can use district-level transaction data to compare banking activity across locations and support decisions about branch services, staffing, and infrastructure allocation.

8.2.7 Banking Operation Amounts

The transaction amount by operation visualization shows that VKLAD and VYBER account for substantial portions of the displayed total transaction amount. Other operations, including PREVOD Z UCTU and PREVOD NA UCET, also contribute to the total.

Recommendation: Management can monitor the amounts associated with each operation to understand the distribution of banking activities and identify areas requiring further operational analysis.

8.3 Overall Recommendations

Based on the dashboard analysis, the following recommendations are proposed:

1. **Monitor transaction trends:** Regularly review yearly and monthly transaction patterns to support operational planning.
2. **Improve resource allocation:** Use regional and district transaction volumes when planning staff, facilities, and system capacity.
3. **Analyse customer behaviour:** Examine transaction patterns across customer groups to understand banking service usage.
4. **Monitor transaction categories:** Track transaction types and operations to identify changes in banking activity.
5. **Improve dashboard monitoring:** Update the dashboard periodically and use appropriate filters to investigate specific years, regions, transaction types, and customer groups.

8.4 Conclusion

The analysis of the Berka banking dataset demonstrates how a Business Intelligence dashboard can support banking decision-making. The visualizations reveal changes in

transaction amounts over time, differences in monthly activity, customer gender distribution, regional variation, district-level transaction volumes, and the distribution of transaction types and operations. These insights can assist management in monitoring banking activities, planning resources, and identifying areas for further investigation.

Use of AI Tools and Acknowledgement

AI tools, including ChatGPT, were used as supporting tools during the development of this Data Warehousing and Business Intelligence assignment. They assisted with understanding data warehousing concepts, improving report explanations, troubleshooting SQL queries and database errors, refining ETL processes, and providing guidance on Power BI dashboard design and data visualization.

All AI-generated suggestions were critically reviewed and adapted to the project requirements. SQL scripts, database structures, data transformations, validation results, and dashboard outputs were checked against the actual implementation. The final report reflects the work performed, the results obtained, and the understanding developed throughout the assignment.

AI tools were used in accordance with the assessment guidelines, following the CLEAR framework for responsible and transparent use of AI.

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